



UNIVERSITY OF
ILLINOIS CHICAGO

DEPARTMENT OF CHEMICAL ENGINEERING

CHEMICAL REACTION ENGINEERING

CHE 321, SPRING 2026



Department of Chemical Engineering
College of Engineering
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CHEMICAL REACTION ENGINEERING

CHE 321, SPRING 2026

I. INSTRUCTOR & COURSE DETAILS

INSTRUCTOR

Korosh Torabi

Email address: korosh@uic.edu

Office Hours: Mondays 1:00-2:00 PM and Tuesdays 12:00-1:00 PM, EIB 246

TEACHING ASSISTANT

Gowtham Vadlamudi

Email address: gvadl@uic.edu

Office Hours: Fridays 11:00 AM-1:00 PM, EIB 249

COURSE BLACKBOARD SITE

<https://uic.blackboard.com>

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Blackboard, email the Learning Technology Solutions team at LTS@uic.edu.

COURSE MODALITY AND SCHEDULE

This course is taught ON CAMPUS.

Tuesdays and Thursdays 4:00 PM - 5:15 PM, EIB Room 124

II. COURSE INFORMATION

CATALOG COURSE DESCRIPTION AND PREREQUISITE/COREQUISITE STATEMENT

Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors. Non-isothermal reactors.

Prerequisite(s): CHE 210; and MATH 220; and Credit or concurrent registration in CHE 301.

BRIEF LIST OF TOPICS TO BE COVERED

Fundamentals of chemical reaction engineering. Design of industrial reactors and bioreactors. Rate laws, kinetics, and mechanisms of homogeneous, heterogeneous, and biological reactions. Analysis of rate data, multiple reactions, and heat effects.

COURSE RATIONALE AND GROWTH MINDSET

Chemical Reaction Engineering (CRE) is central and unique to the chemical engineering discipline. The elements of CRE are utilized in various industries and manufacturing sectors, including petrochemicals, biopharmaceuticals, agrochemicals, polymers, fragrances, flavors, food additives, dyes and chemicals, ceramics, and many more. The

heart of these manufacturing processes is a chemical reaction that converts raw materials or precursors into the molecule of interest. The objective of chemical engineers is to maximize the yield and conversion of the final product in the most energy-efficient and cost-effective manner.

This course will teach fundamental concepts in CRE and provide step-by-step instructions on the evaluation of the performance of the chemical reactor and its optimization. In this course, you will learn how to analyze chemical reactors and reaction systems, develop critical and creative thinking skills, and practice solving reaction engineering problems using computer modeling.

It is strongly suggested that you read the textbook carefully, do the homework problems in detail without consulting other solutions, ask questions in class and during office hours, discuss the material with your classmates, and work on additional problems to master it. Course materials and assignments can be complex and challenging, but they are crucial to intellectual and personal growth and development. There are times when you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, attend office hours, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

COURSES LEARNING OUTCOMES

By the end of this course, students should learn to

- Develop algorithms to solve chemical reaction engineering problems through logic rather than memorization.
- Design isothermal, adiabatic, and non-adiabatic reactors for homogeneous and heterogeneous reactions.
- Analyze multiple reactions carried out both isothermally and non-isothermally in flow, batch, and semi-batch reactors to determine selectivity and yield.
- Determine a rate law from experimental data.
- Describe pathways, mechanisms, and rate-limiting steps of nonelementary, biological, and catalytic reactions.
- Solve reaction engineering problems using computer modeling.

ABET STUDENT LEARNING OUTCOMES

- ABET Student Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- ABET Student Outcome 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- ABET Student Outcome 5: an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
- ABET Student Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

COURSE MATERIALS

REQUIRED TEXTBOOK

Elements of Chemical Reaction Engineering, 6th edition, H. Scott Fogler

Website: <http://websites.umich.edu/~elements/6e>

REQUIRED TECHNOLOGY

Python is a general-purpose language for scientific computing and data processing. Through Jupyter Notebooks, it is used to interact with computational simulations on the web. More information on Python installation and Living Example Problems (LEP) may be found at:

http://websites.umich.edu/~elements/6e/tutorials/Python_tutorials.html

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

GRADING POLICY AND POINT BREAKDOWN

- Homework: 15%
- Term Project: 10%
- Quizzes: 15%
- Midterm Exam I: 20%
- Midterm Exam II: 20%
- Final Exam 20%

ASSIGNMENTS

- **Reading assignments:** Assigned from the textbook. No submission is required.
- **Problem sets (Homework):** submit an electronic copy of your homework on Blackboard.
- **Term Project:** Form teams of up to three students. Each team must provide a report of their work, including the required computer code, and submit it electronically on Blackboard.

POLICY FOR MISSED OR LATE WORK

- **Homework:** Late submissions will not be considered for partial credit. Unreadable and unorganized homework will not be graded.
- **Quizzes:** No makeup will be provided for missed quizzes; however, the lowest quiz grade will be dropped from the final grade calculation.
- **Exams:** No makeup will be provided for missed exams. Students who need to miss an exam for any reason are expected to contact the instructor, preferably in advance of the exam date. With the appropriate documentation, the grade for the missed exam will be calculated based on the average of the other exam grades.

ATTENDANCE AND PARTICIPATION POLICY

Please email me if you face an unexpected situation that may impede your attendance, participation in required class and exam sessions, or timely completion of assignments.

EMAIL EXPECTATIONS

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

ACADEMIC INTEGRITY

As a student and member of the UIC community, you are expected to adhere to the [Community Standards of academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

OTHER COURSE POLICIES

Regrading of exams must be requested within 2 days of the grades being returned. For homework issues, please contact the TA.

IV. COURSE SCHEDULE

Week	Topics	Book Chapter
1	Introduction	0
1	Mole balance and basic reactor models	1
2	Introduction to rate laws, reaction kinetics, and equilibrium	3
3	Reactor mole balance in terms of conversion	2
4	Relating concentrations to conversion	4
4-5	Isothermal reactor design in terms of conversion (single reaction with no side streams)	5
5-6	Isothermal reactor design in terms of concentration Or molar flow rate (membrane reactors and semi-batch reactors)	6
7	Midterm exam 1	1 through 6
8	Isothermal reactor design with multiple reactions	8
9-10	Energy Balance and non-isothermal reactor design in terms of conversion (adiabatic reactors)	11
11	(March 23-27) Spring break. No classes.	
12	Midterm exam 2	8 and 11
12-13	Energy Balance and non-isothermal reactor design in terms of concentration or molar flow rate	12
14	Reaction kinetics data collection and analysis	7
15	Reaction pathways and bioreactors	9
16	Introduction to catalysis	10
17	Final exam	Comprehensive

- The due dates for homework assignments and term projects will be posted on the Blackboard site.
- Quizzes will be scheduled during regular class time and location with at least two days' notice.
- Midterm exams will take place at the regular class time and location.
- The final exam will be held on the date and time specified by the university's Registrar's Office.

DISCLAIMER

This syllabus is intended to provide students with guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make

changes as needed to address course requirements. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

V. ACCOMMODATIONS

DISABILITY ACCOMMODATION PROCEDURES

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc@uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodation, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

RELIGIOUS ACCOMMODATIONS

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: "YOUR NAME: Requesting Religious Accommodation."

VI. CLASSROOM ENVIRONMENT

INCLUSIVE COMMUNITY

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

NAME AND PRONOUN USE

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are he/him. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

COMMUNITY AGREEMENT/CLASSROOM CONDUCT POLICY

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.

- Debate concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.

CONTENT NOTICES AND TRIGGER WARNINGS

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

VII. RESOURCES: ACADEMIC SUCCESS, WELLNESS, AND SAFETY

We all need the help and the support of our UIC community. Please visit my office hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

ACADEMIC SUCCESS

- UIC [Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#)
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)

WELLNESS

- Counseling Services: You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- Campus Advocacy Network: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at <http://can.uic.edu/>.

SAFETY

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.